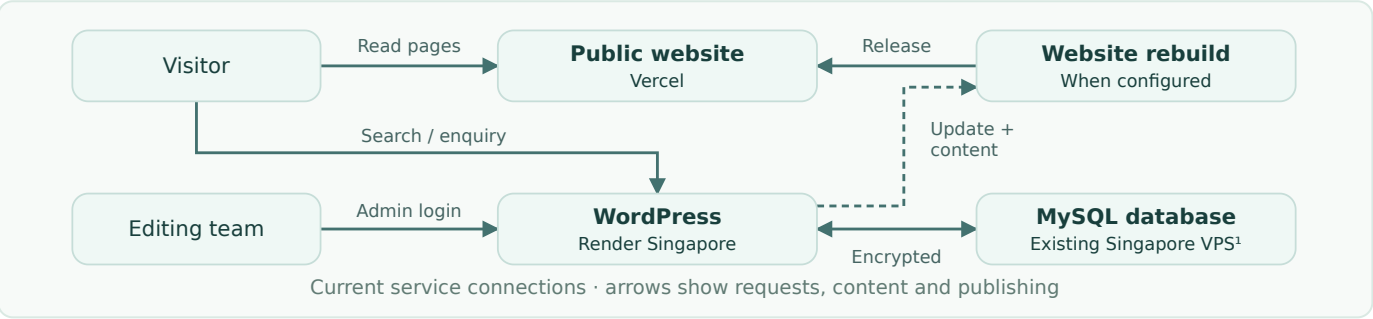
CareConnect: taking the directory to production

Planning brief: 20,000 monthly visits · five editors · data kept in Singapore.
Reviewed: 28 October 2026. This concise design separates existing features from proposed production work. The owner reports that the three hosted services are running; private settings and recovery arrangements have not been independently audited.

1. Architecture — how visitors reach the content

CareConnect combines a Next.js public website, a WordPress editing area and a MySQL database holding content, accounts and private enquiries. The **CareConnect Services** plugin supplies the directory features.



Current service connections · arrows show requests, content and publishing

¹ A VPS is a rented server. Confirm its Singapore location in the hosting account.

Visitor journey: reading a page usually needs only the public website. Search asks WordPress for published results. Enquiries are stored privately before email is attempted; demo email is best-effort. Visitors never access the database directly. Existing pages can remain readable during a WordPress outage, while search and enquiries fail.

Content updates: saving changes updates the database immediately; public pages change after a successful rebuild. The plugin can request this through a configured Vercel deployment link. Recent testing found updated WordPress content without a new deployment, so this connection needs confirmation. The build command, `npm run build`, clears saved content responses before generating pages.

CMS choice: WordPress offers familiar editing, draft review and individual permissions, with directory rules kept together in the plugin. **Sanity was rejected** because it would replace the required WordPress editing experience and require a different integration.

2. Content model — what the team manages

Content	Main fields
Service	Name, type, short introduction, full description, eligibility, address, phone, hours, image link/alternative text, review date, up to three related services, optional map location
Article	Title, summary, full text, category, publication date, image link/alternative text
Home page	Heading, supporting sentence, ordered featured services; latest articles appear automatically
Private enquiry	Sender name, email and message; administrators only

Short introductions and full descriptions are separate fields. Required-field checks keep invalid services as drafts. Reviews older than twelve months are flagged. Images use links or bundled illustrations; uploads are disabled because demo file storage is temporary.

Editor **Draft** → Submit **Review** → Publisher **Approve** → Website **Rebuild** → Visitors **See update**

Give the five editors individual accounts. Editors manage their own unpublished work; designated publishers review, publish and maintain home content. Administrators manage settings and private enquiries. Drafts are excluded from public responses. Previously published pages can remain visible until the website is rebuilt.

3. Hosting — locations, environments and releases

Current: Vercel serves public pages, Render's configuration selects Singapore for WordPress, and an existing VPS hosts MySQL. Local development uses Docker. Separate staging and automated release checks remain proposed.

Residency gap: Vercel caches pages globally, so the current setup cannot promise every hosted copy stays in Singapore. Selecting a Singapore function region would not change this static website's delivery. See Vercel's delivery-network documentation.

Production proposal: retain the application and content model; use paid, always-on Render WordPress in Singapore, dedicated MySQL and static-website servers in DigitalOcean Singapore (`sgp1`), and encrypted Singapore backup storage. Replace the Vercel-specific trigger with a reliable Singapore build job. These changes are not implemented. Singapore is available on Render and DigitalOcean.

Keep content-bearing builds, logs and backups in Singapore. Verify provider support access, email processing and backup locations before approval. Disable external maps and unapproved image hosts pending review. This scope covers provider-held copies; visitors still receive requested public pages.

Environment	Purpose and separation
Development	Local Docker, fictional data, no production enquiries
Staging	Private Singapore test site; separate database, accounts, secrets and update trigger
Production	Live Singapore services, restricted access, scheduled backups

Proposed release path: Git change → automated checks → packaged WordPress release and website build → staging review → approved production release. Build live content on a Singapore worker. Retain previous releases for rollback without overwriting newer enquiries. Currently, Render uses the repository's Docker configuration; Vercel builds from `web/` with `npm run build`.

4. Security — protecting visitors and the team

Area	Existing protection	Production addition
Connections	HTTPS hosting; supplied setup configures database encryption	Verify live controls and database certificate identity; restrict database access to approved servers
Admin	Separate editor, publisher and administrator permissions	MFA: a second login check; remove demo access and protect the admin entrance
Malicious content and forged edits	Validate entries, limit formatting, safely display text, use WordPress's structured database queries, check editing permissions and verify admin forms	Maintain security updates and repeat access checks after releases
Spam and automated attacks	Hidden bot field, timing checks, submission and size limits	Login throttling, firewall rules and flood protection; verify visitor-address handling behind hosting proxies
Secrets and backups	Passwords and deployment links stay outside public code	Rotate credentials; encrypt backups with separately restricted access

Logs must omit enquiry text, emails and secrets. Agree retention before launch; a starting proposal is 90 days for enquiries and 30 days for logs. Backup expiry and deletion procedures must follow that policy.

5. Operations — detecting problems and recovering

Assign a technical owner and backup contact. Check pages and search every five minutes. Alert on sustained failures, low database memory/disk space, failed backups, failed rebuilds and email errors. Verify enquiry storage using controlled fictional submissions and cleanup. Keep restricted troubleshooting logs in Singapore.

Take an encrypted, consistent database backup nightly and before risky changes. Retain 30 daily copies in separate Singapore storage, plus versioned application releases and website exports. Check completion daily; practise restoration quarterly in isolation.

Proposed target	Client meaning
RPO: 24 hours	A major failure could lose up to one day of edits and enquiries
RTO: 4 hours	Aim to restore service within four hours of declaring an incident

Restore the database and compatible application, check accounts and private-content protection, rebuild pages, then test search and enquiries. These targets are not demonstrated guarantees: they require an available responder. A Singapore-wide outage may take longer because overseas recovery copies are excluded.

6. Cost — estimated monthly budget

Estimates, verify; USD/month. Assume 20,000 visits, four pages per visit and 0.5 MB per page: approximately 40 GB of public delivery. Allow five WordPress accounts, modest storage, one production instance per component and one staging server. Peak capacity still needs testing.

Item	Monthly allowance
Singapore public-website server, 1 GB memory	$6
Always-on Render WordPress, 2 GB memory	$25
Dedicated Singapore MySQL server, 2 GB memory	$12
Separate staging server, 2 GB memory	$12
Singapore backup storage	$5
Monitoring, email and usage contingency	$5–20
Estimated total	**$65–80**

Baselines checked on 28 October 2026: DigitalOcean servers, Render, backup storage. The contingency is a planning allowance. Backup storage must be private with global delivery disabled. Excludes tax, domains, engineering/on-call labour and redundant standby servers. Editor accounts do not require five hosting-dashboard seats. The current demo uses free tiers plus an existing VPS; its bill has not been reviewed.

7. Risks and next steps — another working week

The shared 1 GB VPS has already experienced a MySQL memory failure. Free Render services can sleep. Failed rebuilds leave old information visible; rapid edits within the plugin's one-minute trigger window can be skipped without a retry queue. Live search may find services before their pages are rebuilt. Urgent withdrawals need an explicit removal process.

When	Deliverable
Days 1–2	Confirm residency and provider terms; provision staging and dedicated capacity; enable MFA and verified database encryption
Days 3–4	Add monitored, retryable publishing; rehearse urgent withdrawal; schedule backups and demonstrate restoration
Day 5	Test peak traffic, editorial roles and enquiries; review alerts, retention and release checks with the client

A second language, image uploads, guaranteed email delivery and automatic failover remain outside the implemented scope. Approve production after evidence of a successful publishing cycle, recovery drill and residency review.